

Summary: Visualizations In Power BI - I

SESSION OVERVIEW:

By the end of this session, students will be able to:

- Get comfortable with visualizing measures, columns and metrics.
- Visualize data in Power BI using bar charts, column charts, combination charts etc.
- Visualize metrics using Card and apply filters using slicers.

KEY TOPICS AND EXAMPLES:

Note: For this session, we will be using [this dataset](#). In the beginning, it is better to use DAX to add a Sales column to this dataset, which would be used for visualizations in this dataset.

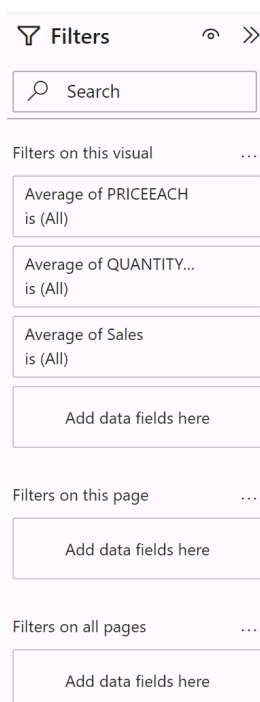
Till now we have seen how to clean and transform data in Power Query editor and how to transform data and perform calculations using DAX. Now we will move to the next step in the Data Analytics Lifecycle: **Data Analysis** and **Visualization**.

1. Visualizations in Power BI

Power BI provides various kinds of charts/graphs/visuals for visualizing our data and identifying trends, patterns, insights etc. When presenting your data to stakeholders, visualizations are powerful tools that can be used to build a story. There are many kinds of visuals like charts, graphs, maps, slicers etc.

The 3 Panes in Power BI Report View:

1. **Filters:** It is used to apply filters to the visuals. It has 2 types of filters: 'Filters on this page' and 'Filters on all pages'.
Also, if we select a visual, another type of filter appears: 'Filters on this visual'. All 3 filters can be seen below:



At the bottom of the Power BI Desktop Report View there is an option to add a page (+ symbol).



- If we apply a filter in the 'Filters on all pages' section, it will create a filter across pages.
- If we apply a filter in the 'Filters on this page' section, it will create a filter for all filters on this page and only for this page.
- If we apply a filter in the 'Filters on this visual' section, it will create a filter only for the selected visual.

2. Visualizations: It provides options for adding visuals, defining the values to be displayed in the visual and formatting the visual. It has 3 tabs: 'Add data to your visual', 'Format your visual' and 'Add further analyses to your visual'.

3. Data: It has the list of tables and the fields/columns in each table. To create a visual, we can simply drag a field from the Data pane to the Report View area.

Types of visualizations in Power BI:

A. Slicer

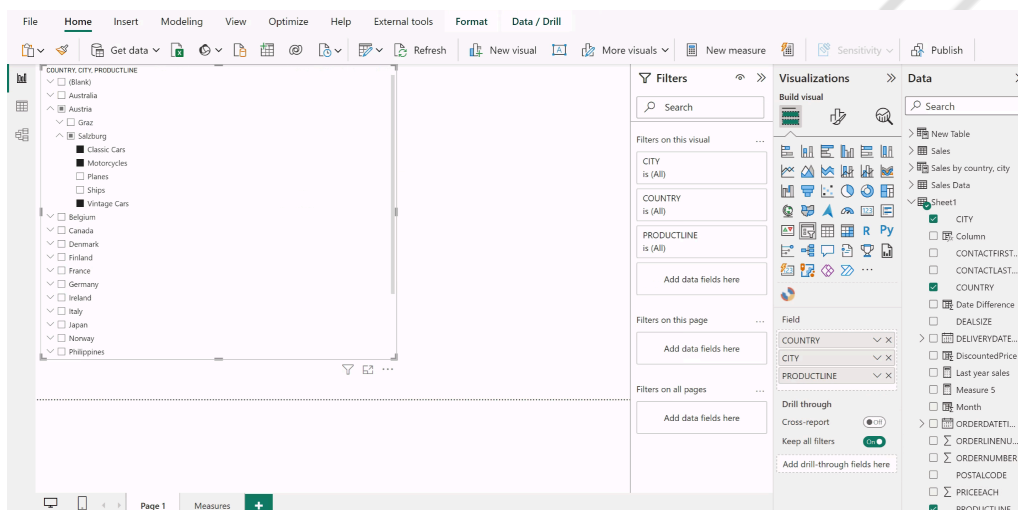
It helps us filter the data based on a column.

In our [Vehicle Orders dataset](#), we can filter using City, Country and Product Line. For adding a field, we can simply check the box next to the field or drag and drop the field in the 'Field' box in the Visualizations pane.

In this case we can either create a single slicer or create 3 different slicers - one for each field (recommended). Both are shown below.

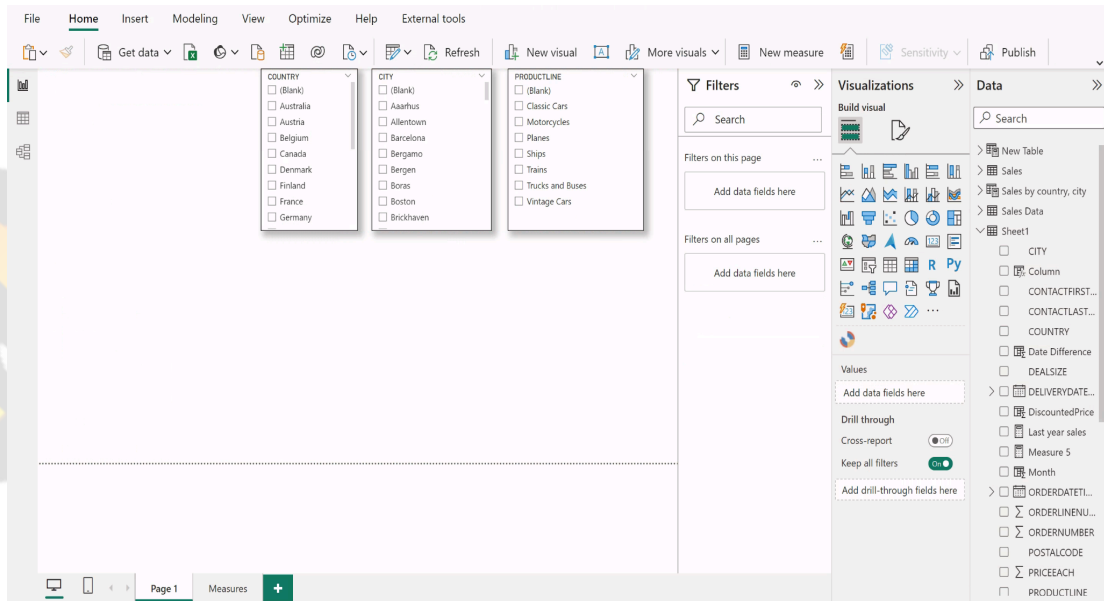
The Single Slicer:

If we want to create a single slicer and select Country: **Austria**, City: **Salzburg** and Product line: **Classic Cars**, **Vintage Cars** and **Motorcycles**, we can do this as shown in the following image.



The 3 different Slicers:

Create one slicer for each of the 3 fields: Country, City and Productline.



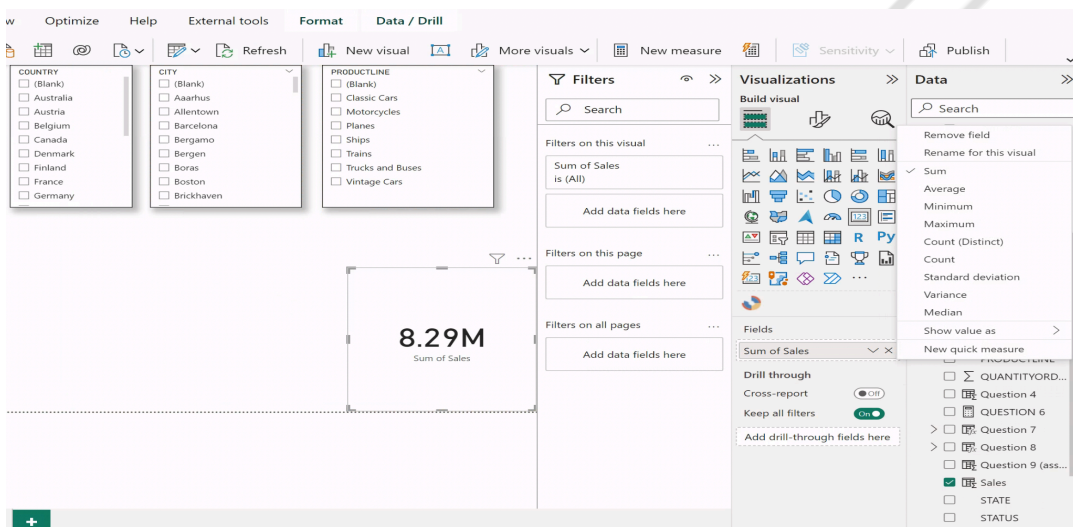
In particular, slicers for Date are used often. In our dataset, we can create a slicer for a date column like ORDERDATETIME, as shown below.



B. Card

It is typically used to display a value like a measure or a calculation. Drag the value in the Fields section in the Visualization pane and select the aggregation as shown below. You can also choose to not aggregate using the 'Don't Summarize' option if available.

In our dataset, we intend to see the **sum of sales**. We can right click on the field that has been added to the Fields section and select Sum.



Note: Sales column can be added as a calculated column using DAX:

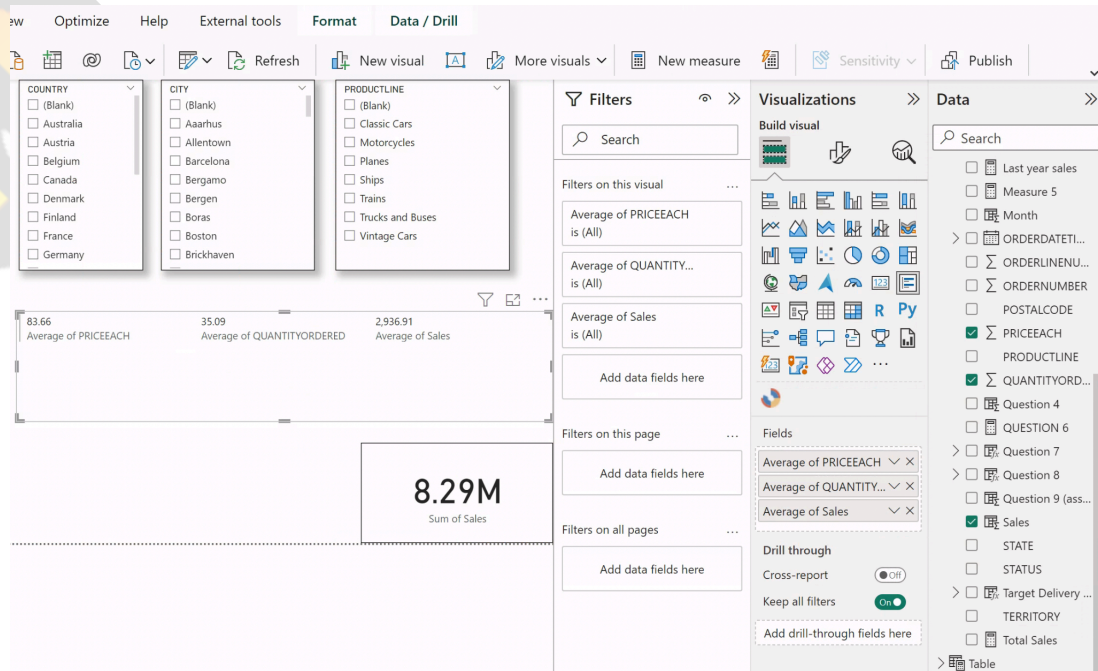
$Sales = Sheet1[QUANTITYORDERED] * Sheet1[PRICEEACH]$

C. Multi-row Card

It displays multiple values within the same card.

In our dataset, we intend to see Average Price, Average quantity ordered and Average Sales.

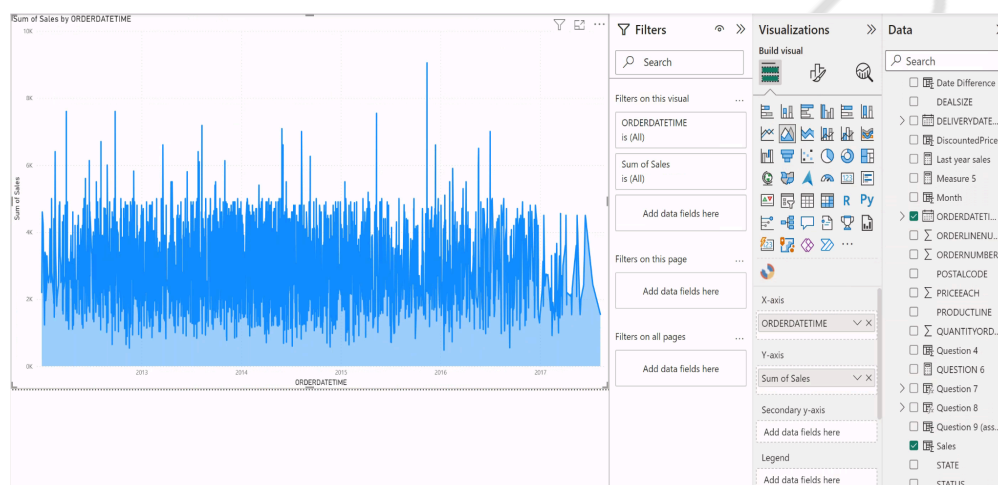
We can do it as shown below:



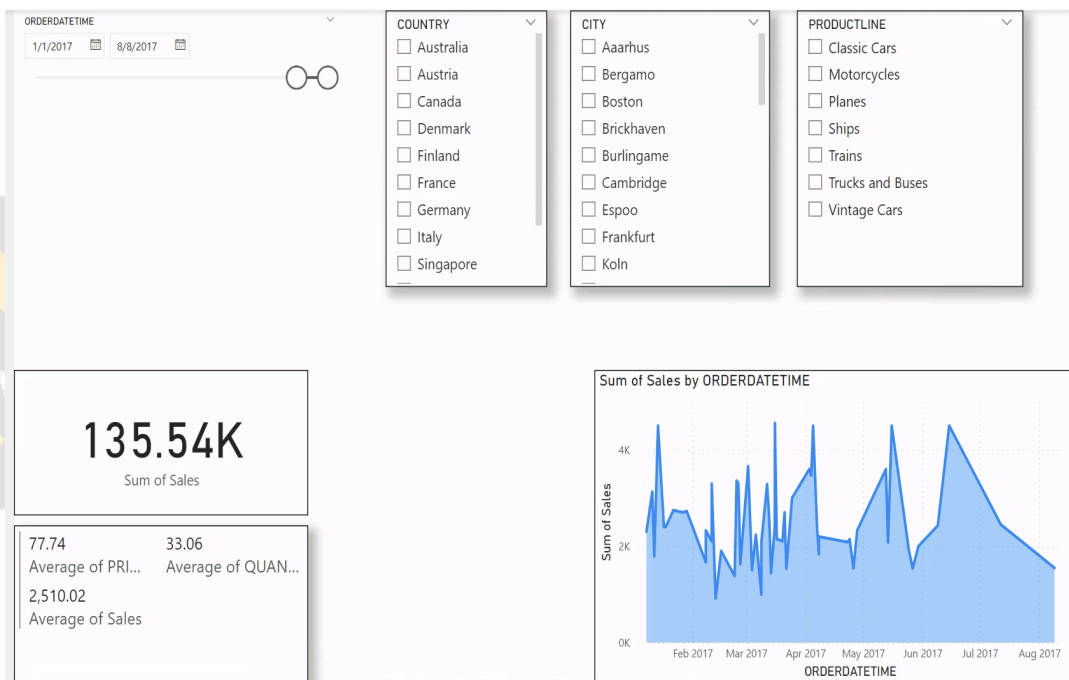
D. Area Chart (Normal Area Chart)

In an area chart, the total value of a field is shown as area under the chart. It can be used to show the magnitude of a field over time. It can be used to compare trends across time.

In our dataset, we can show total sales over time using the Area chart. This can be done by keeping ORDERDATETIME in the X-Axis section and taking the Sum of Sales in the Y-Axis section of the Visualizations pane. The series used here is Sales.

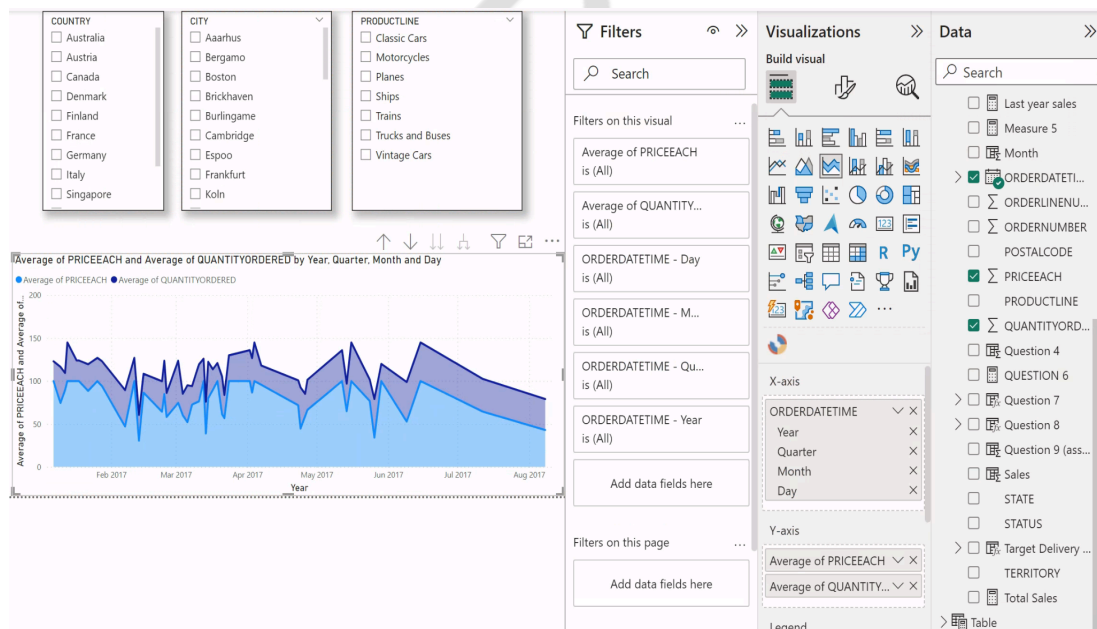


If we see only for 2017, the area chart will look like this:



Area charts are of 2 types: Normal area chart and Stacked area chart.

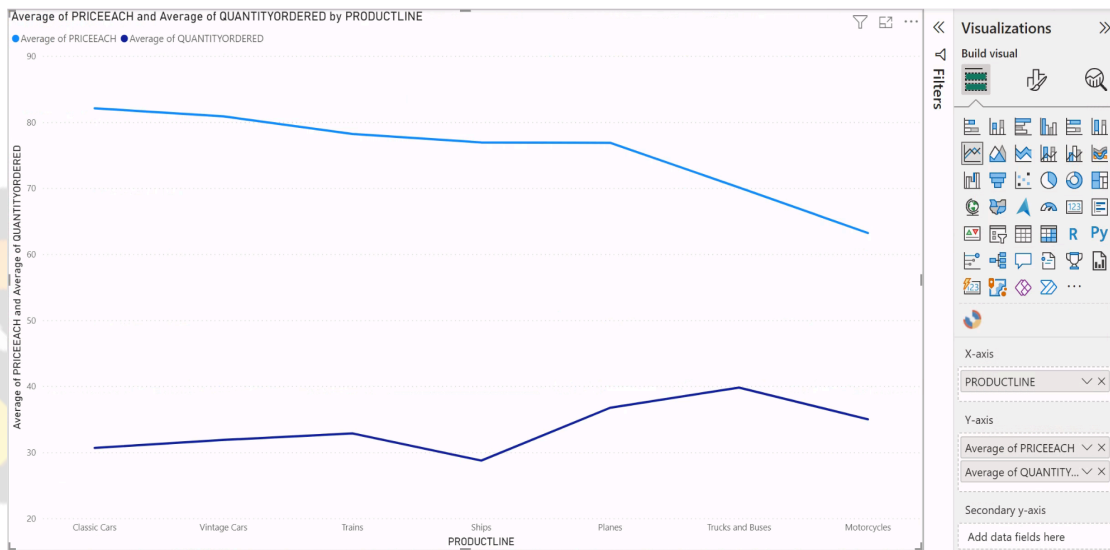
Stacked Area Chart: It stacks each series on top of the other. If we want to visualize both Average Price and Quantity in the same chart with stacking, it can be done as shown below.



E. Line Chart

A line chart shows the trend of a series. Data points are plotted on the chart and are connected by straight lines. More than one series can be plotted in a line chart.

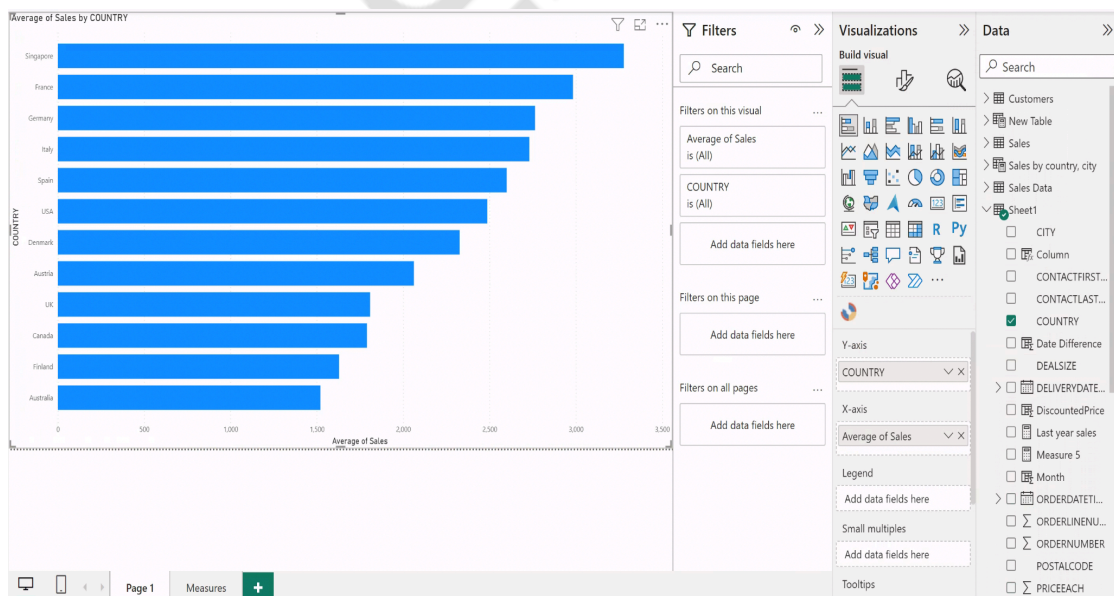
In our dataset, we can see the trend of average price and quantity ordered by product line.



F. Bar Chart

It is used to visualize values in various categories: bars. A bar chart has bars in horizontal orientation.

Bar chart for average sales by country:

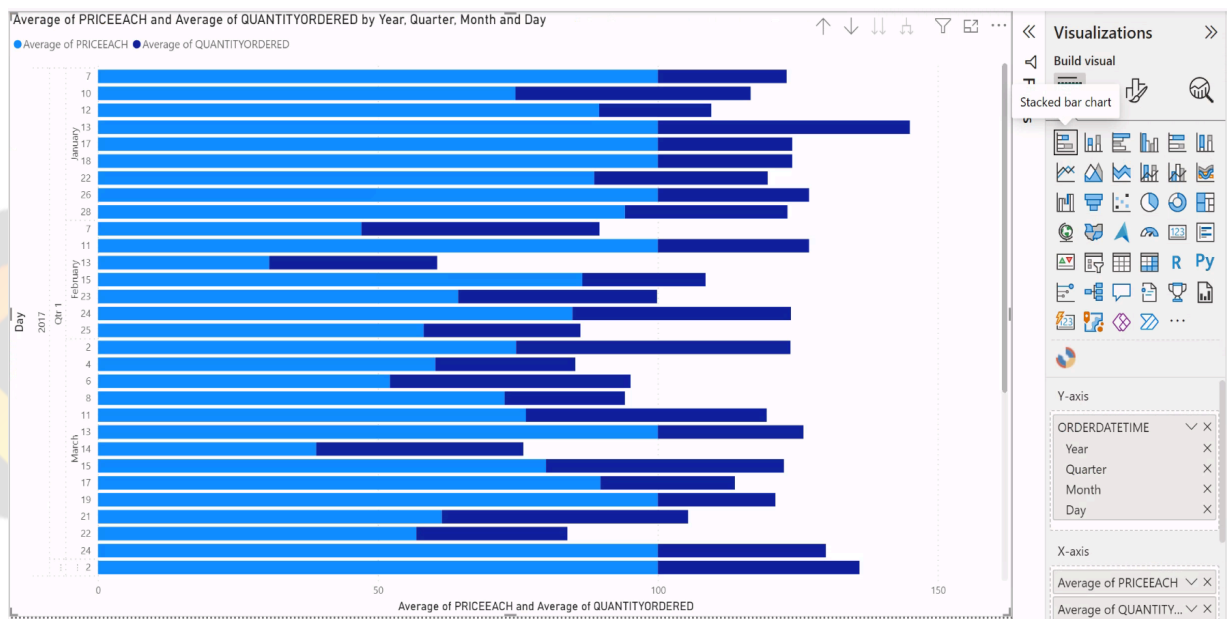


There are 3 types of bar charts: Stacked, 100% Stacked and Clustered. This distinction is based on how more than one series is presented in the chart.

Stacked chart

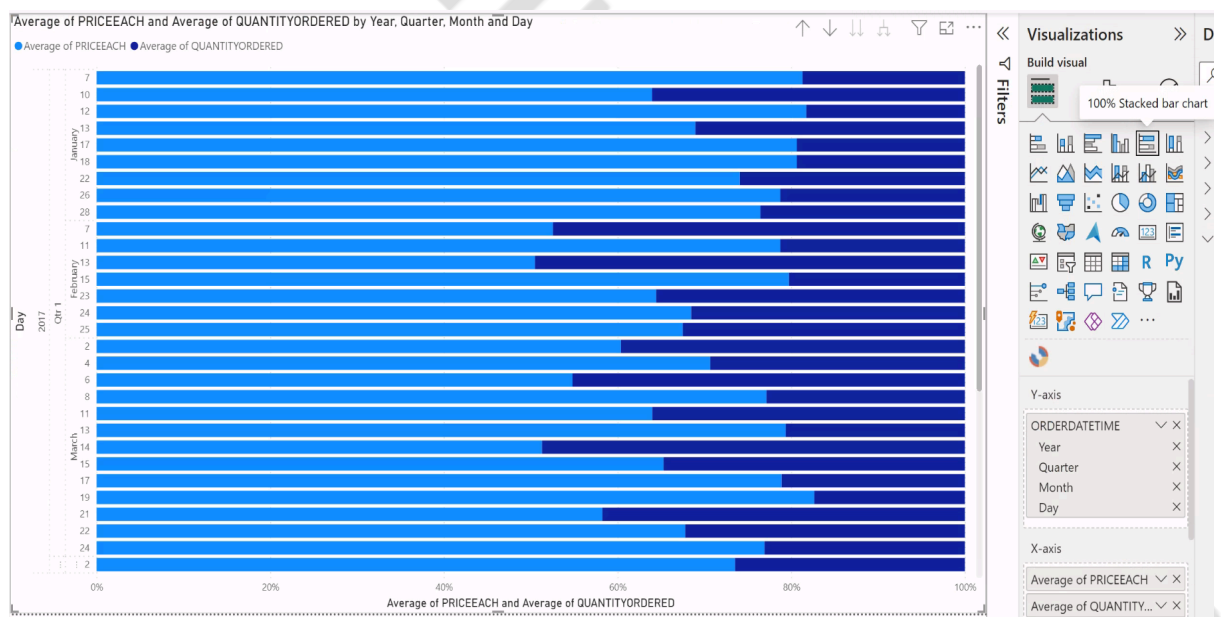
We can visualize the average of price and quantity ordered in bars stacked on each other horizontally. Each series is represented by a different colour.

We can visualize the Avg Price and quantity in the same chart using the stacked chart:



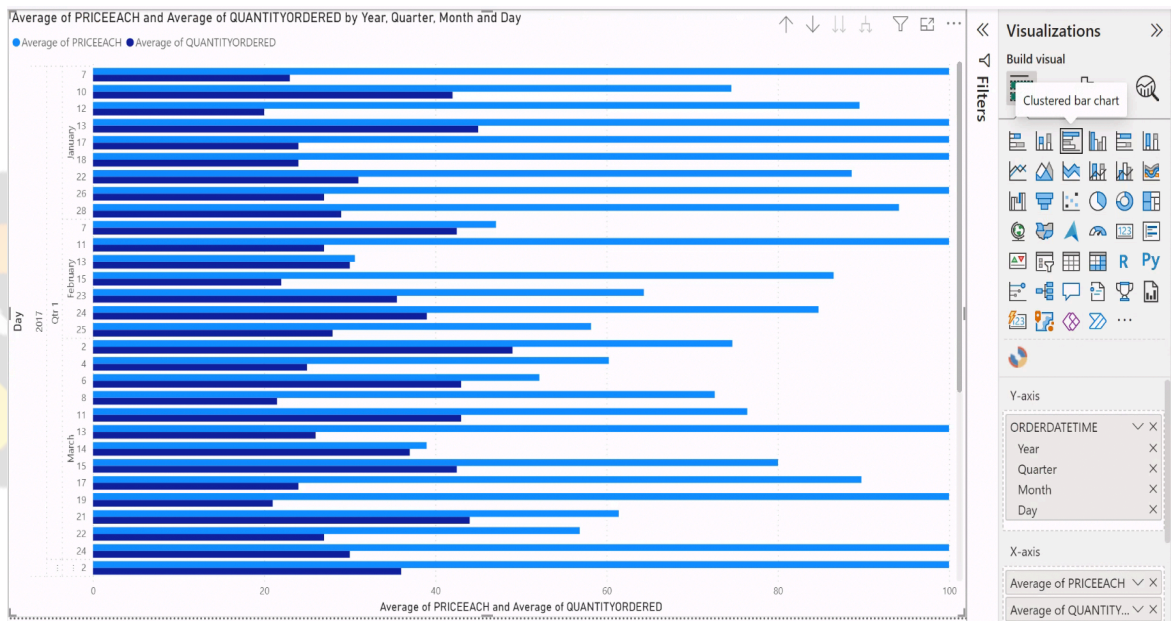
100% Stacked chart

In the 100% stacked chart, the stacking is 100% for every column. We can visualize the average of price and quantity ordered in bars stacked on each other horizontally.



Clustered chart

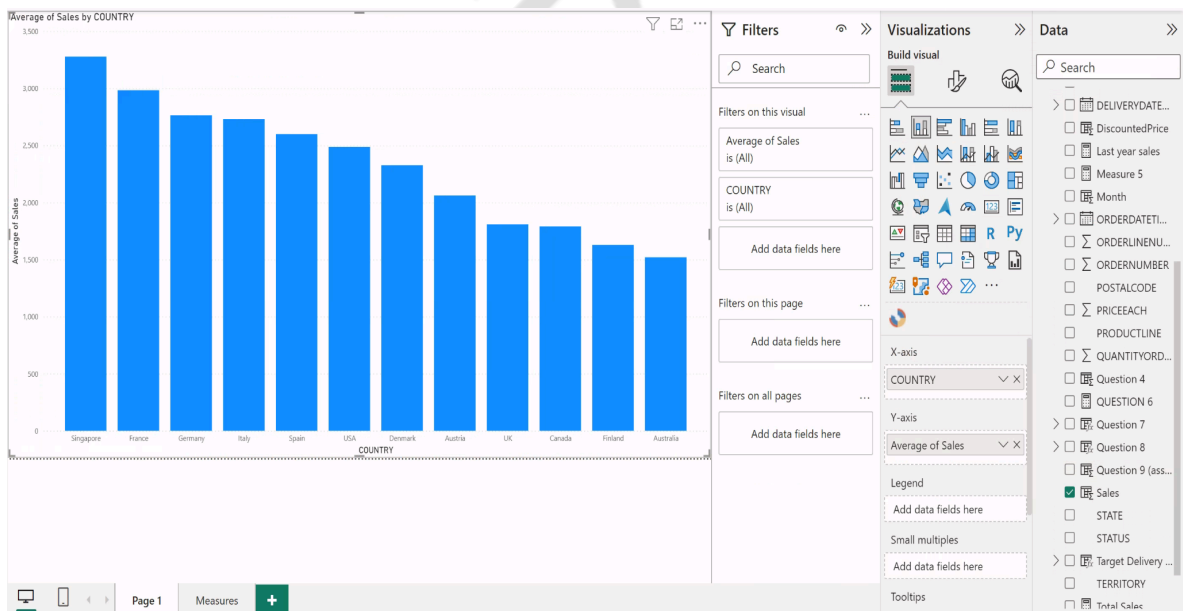
It creates a bar for each series and bars for the same category on the x-axis (here, Date) are clustered near each other.



G. Column Chart

It is used to visualize values in various categories: columns. In the column chart the bars are oriented vertically.

Column chart for average sales by country:

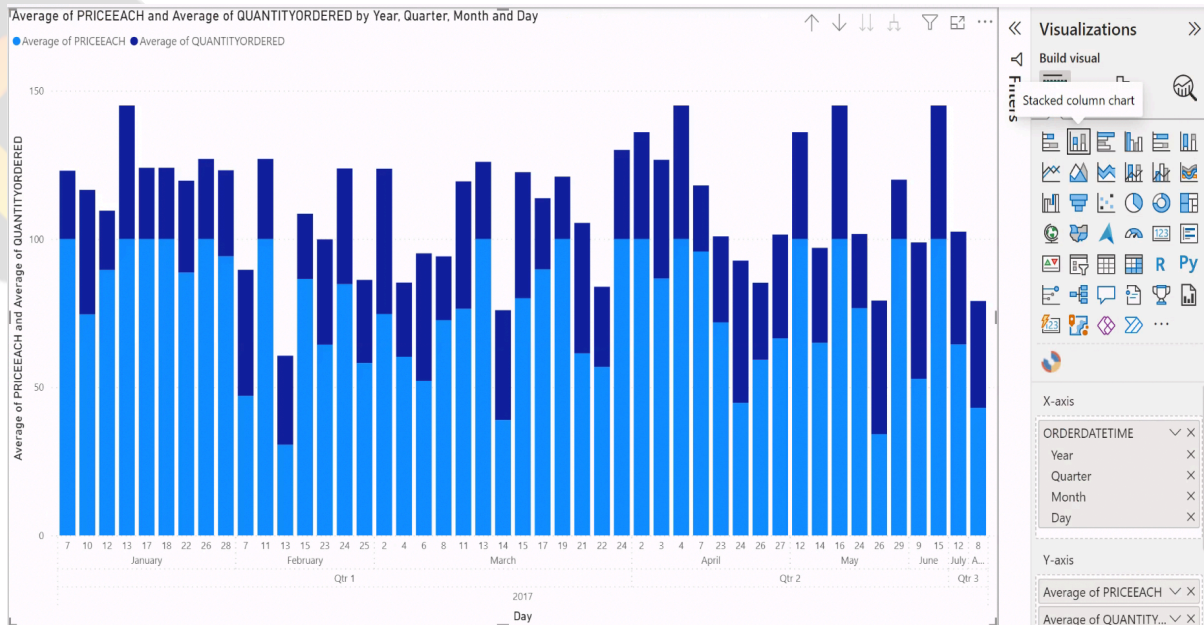


There are 3 types of bar charts: Stacked, 100% stacked and Clustered. This distinction is based on how more than one series is presented in the chart.

Stacked chart

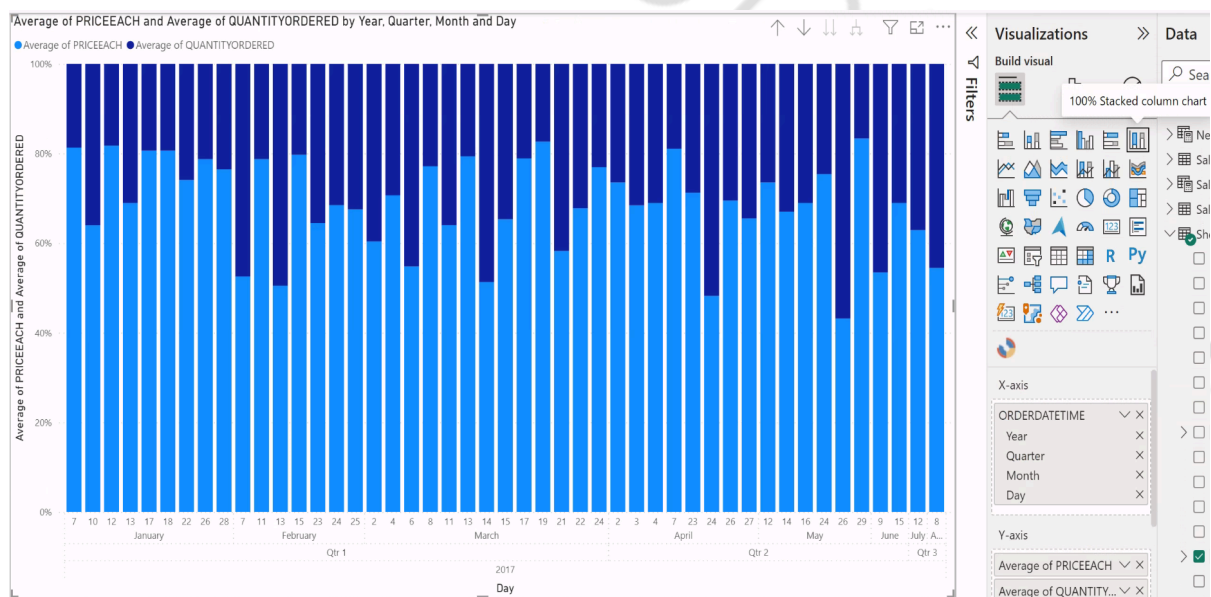
In this type of chart, the columns are stacked on top of each other vertically. Each series is represented by a different colour.

We can visualize the Avg Price and quantity in the same chart using the stacked chart:



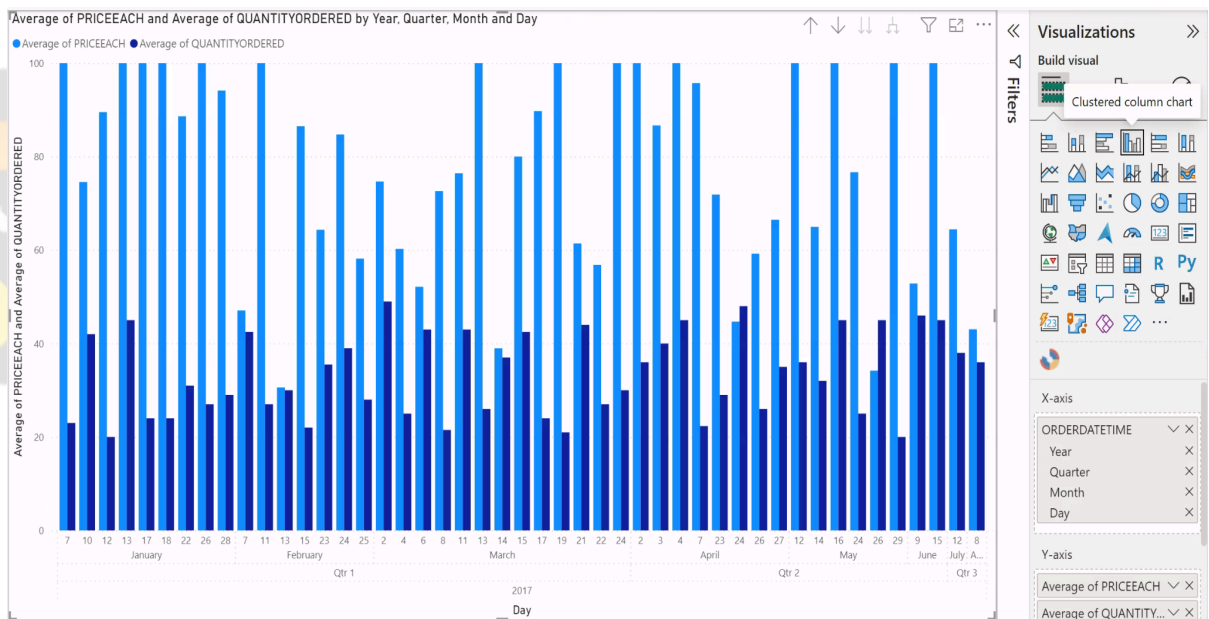
100% Stacked chart

In the 100% stacked chart, the stacking is 100% for every column. We can visualize the average of price and quantity ordered in bars stacked on each other vertically.



Clustered chart

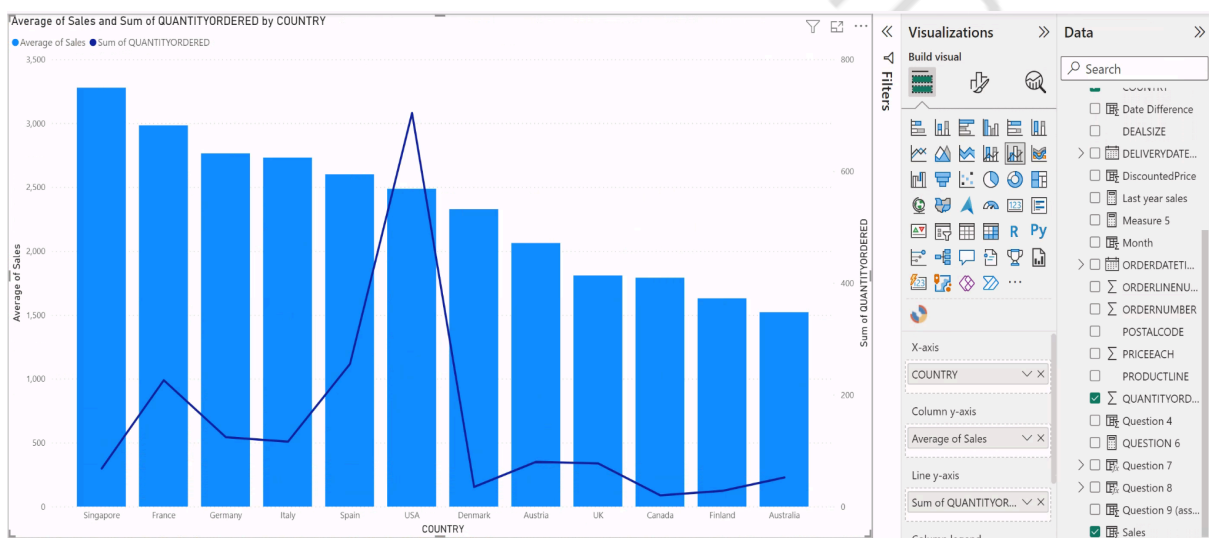
In a clustered chart, there is a column for each series and columns for the same category on the x-axis (here, Date) are clustered near each other.



H. Combination Chart

It is used to visualize multiple fields in multiple formats in a single visualization. The X-axis is shared but the Y-axis is different for the two fields. The y-axis legends appear on either side of the visualization.

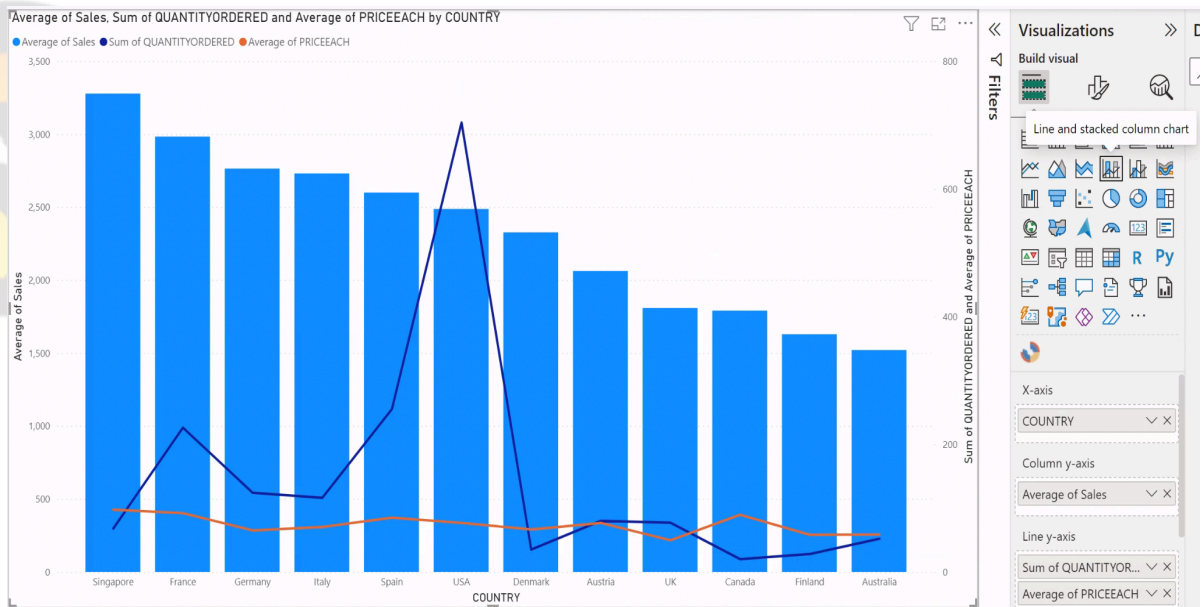
In our dataset, we can create a combination chart as shown below. We can visualize both the average sales (has a high scale of values, in thousands) and the total quantity ordered in each country (has a low scale of values, in hundreds) on the same chart, using the same x-axis (Country) but different y-axes.



There are 2 types of clustered charts available in Power BI:

- **Line and stacked**

‘Average sales’, ‘average price of item’ and ‘total quantity ordered’ by country:



- **Line and clustered**

‘Average sales’, ‘average price of item’ and ‘total quantity ordered’ by country:

